Does 98 % Survive Unseen Patients? A Code-Exact Reproduction and Patient-Level Ablation of an EfficientNet-B0 Ensemble for Malaria Cell Classification

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Code, results and trained models: https://github.com/amathziah/malaria-detection

***Abstract*—Deep-learning studies on the NIH malaria cell images report 96–99.9 % accuracy, mostly with random image-level splits that place cells of the same patient in both the training and the test set. We reproduce the EfficientNet-B0 ten-fold ensemble of Marques *et al.* (2022) from the authors' executed code, then test with a pre-registered ablation whether its accuracy survives patients the model has never seen (H1) and whether YUV histogram-equalisation stain normalisation helps on them (H2). The code-exact reproduction lands within 1 percentage point (pp) of the paper on every metric (ensemble 97.62 vs. 98.29 %; mean fold 97.55 vs. 97.56 %). It also shows that the paper's text swaps precision and recall and that its evaluation images were augmented. On 40 held-out patients the ensemble reaches 98.45 % accuracy (95 % patient-bootstrap CI 97.94–98.91 %), so H1 is not supported by its pre-registered test. Scored on identical cells, models that had seen other cells of the same patient gain only 0.85 pp of Matthews correlation (MCC; 95 % CI 0.05–2.02) across 160 patients, and nothing on the 40 held-out patients: patient leakage is real but small. YUV normalisation does not help on unseen patients: its effect lies within $\pm$1 pp of MCC (90 % CI -0.73 to $+0.16$). In an exploratory patient-level analysis, a rule chosen on the 160 cross-validation patients identifies 26 of 28 infected and all 12 healthy held-out patients. Each of the 40 models trains in about 11 minutes on an RTX 4090 cloud GPU.**

***Index Terms*—malaria, thin blood smear, EfficientNet, ensemble learning, reproducibility, data leakage, patient-level validation, stain normalisation**

I. Introduction

Malaria is diagnosed by examining Giemsa-stained blood smears under a microscope, where a trained technician inspects red blood cells for parasites. The method is the clinical gold standard, but it is slow, it depends on scarce expertise in exactly the regions where malaria is common, and its results vary between observers and laboratories. Automated classification of single red-blood-cell images is therefore an active research topic, and on the public NIH Malaria Cell Images dataset [2] convolutional neural networks (CNNs) routinely report accuracies of 96–99.9 % [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11].

Two questions decide whether such numbers can be trusted. First, *can they be reproduced?* Papers rarely publish their full training code, and small undocumented choices (augmentation, label smoothing, checkpointing) move the results by more than the differences being claimed. Second, *do they hold on patients the model has never seen?* Most studies split the 27,558 cell images at random, so cells of the same patient—sharing stain, illumination and the patient's own blood—appear in both the training and the test set. A model can then score well by recognising the patient rather than the parasite.

This work addresses both questions for a strong, openly documented baseline: the EfficientNet-B0 ten-fold ensemble of Marques *et al.* [8], whose authors published their executed notebook. Our contributions are:

1) A *code-exact* reproduction of [8] on all ten folds, built from the authors' notebook rather than the paper's text, verified by automated tests against the paper's printed outputs, and trained in about 11 minutes per fold instead of 4 h 46 min.
2) Three corrections to the paper's reporting: its text swaps the ensemble's precision and recall, its evaluation images were randomly augmented, and one table reports $(\text{precision} + \text{recall})/2$ as accuracy.
3) A pre-registered ablation that changes one variable at a time: a patient-level split (hypothesis H1: random splits inflate the scores) and YUV histogram-equalisation stain normalisation [4] (hypothesis H2: it helps on unseen patients), with a control that applies the normalisation under the original split.
4) A patient-level statistical analysis—cluster bootstrap over patients, corrected resampled t-tests [14] and Holm correction [15]—with the hypotheses, decision rules and primary metric committed to the repository before any of the new models was trained.
5) Exploratory analyses fixed in a dated addendum before they were run: equivalence bounds, leakage measured on identical cells, a patient-level diagnosis rule chosen without the test patients, and an error audit.

II. Related Work

Liang *et al.* [1] showed that a 16-layer CNN trained from scratch (97.37 %) outperforms transferred AlexNet features (91.99 %). Rajaraman *et al.* [2] released the NIH dataset and mapped which layers of pre-trained CNNs transfer best;

importantly, they evaluated at the patient level as well as the cell level and observed lower scores at the patient level. Ensembles followed: Rajaraman *et al.* [3] combined VGG-19 and SqueezeNet, and Marques *et al.* [8] averaged ten EfficientNet-B0 models from a ten-fold cross-validation (98.29 % accuracy, ROC-AUC 99.76 %). Lightweight models target point-of-care use: Fuhad *et al.* [5] reached 99.23 % with about 4,600 FLOPs on a phone after cleaning the dataset's labels, and Masud *et al.* [7] used a 409 K-parameter CNN. Zhao *et al.* [6] extended detection to a second species. Umer *et al.* [4] preprocessed images by YUV conversion and luma histogram equalisation before a stacked CNN and reported ∼99.9 %. Capsule networks [9] and systematic backbone comparisons [10], [11] complete the picture.

Two gaps run through this literature. Protocols differ (5-fold, 10-fold and 80:20 splits; original or cleaned labels), so reported accuracies are not directly comparable. And most studies split images at random, so the same patient's cells sit on both sides of the split; only Rajaraman *et al.* [2], [3] report patient-level evaluation. We take the strongest reproducible baseline and measure how much of its accuracy survives a patient-level split.

III. Data

The NIH Malaria Cell Images dataset [2] contains 27 558 segmented single-cell crops (13 779 parasitized, 13 779 uninfected) from Giemsa-stained thin smears of 200 patients (150 infected with *P. falciparum*, 50 healthy), photographed through a microscope with a smartphone at Chittagong Medical College Hospital, Bangladesh. Crops vary in size and are resized to 224×224.

Patient identity. Every file name begins with the patient's case number, e.g. `C33P1thinF_IMG_..._cell_179.png` belongs to patient C33. We verified this key against NIH's own patient-to-cell mapping files: all 27 558 cells map, giving 200 patients. NIH lists one patient (C47) under two keys, one per microscope; the case number correctly merges them. Infected patients also contribute uninfected cells—10 364 of the 13 779 uninfected cells come from infected patients' smears—so under a random split almost every patient has cells on both sides. Patients differ widely: 65 to 702 cells each, and 1 to 90 % parasitized among the infected.

IV. Methods

A. Code-exact reproduction of the baseline (A0)

The supplementary file of [8] is the authors' executed notebook. It differs from the paper's text in several respects, and we follow the code: noisy-student EfficientNet-B0 weights [12], [13] (qubvel `efficientnet` 1.1.1, all layers trainable) followed by global average pooling and dense layers of 128, 64 and 32 units (ReLU, dropout 0.3) and a two-way softmax; categorical cross-entropy with label smoothing 0.1; Adam (learning rate 10^{-4}); learning-rate halving on a validation-loss plateau (patience 6, minimum 10^{-6}); 33 epochs of batch size 16 with the best-validation-loss weights restored; and a heavy `albumentations` pipeline (flips, 90° rotations, noise, blur, shift-scale-rotate, optical, grid and piecewise-affine distortions, CLAHE, sharpening, embossing, contrast and brightness changes). The data are split 80/20 into a learning set and an untouched hold-out set; a stratified ten-fold cross-validation on the learning set yields ten models, and the ensemble averages their softmax outputs on the hold-out set.

TABLE I
Ablation design. Every experiment uses A0's recipe and ten folds.

ID	Split	Preprocessing	Question
A0	image (paper)	none	baseline
A1	patient	none	H1: leakage (A1 vs A0)
A2	patient	YUV + hist.-eq.	H2: normalisation (A2 vs A1)
A3	image (paper)	YUV + hist.-eq.	control (A3 vs A0)

We reimplemented this recipe in TensorFlow 2.15 with the same Keras 2 API and library versions, and wrote 32 tests that compare it with the paper: the parameter counts of its Table 2 (4 223 934 total, 4 181 918 trainable), the split and fold sizes printed by its code, the exact augmentation pipeline, image loading identical to the authors' generator, the loss, optimiser and callbacks, and its classification reports. The paper's split was unseeded and cannot be recovered; we sort the file list and seed the split so that every machine builds the same one, and store a fingerprint of the split with every fold.

Speed without changing the computation. Images are decoded once into a shared `uint8` array; the augmentation runs in worker processes, each batch seeded by (fold, epoch, batch) so results do not depend on the number of workers; batches reach the GPU as `uint8` and are scaled inside the model; and the training step is XLA-compiled. Arithmetic remains float32. Each fold trains on its own RTX 4090 in about 11 minutes, against 4 h 46 min per fold reported for a GTX 1050.

B. Ablation design (A1–A3)

Each experiment changes one thing relative to A0 (Table I); everything else, including the per-fold seeds, is identical.

Patient split (A1, A2). `StratifiedGroupKFold` with five groups proposes five hold-out sets of about 20 % of the cells; we take the one closest to 20 % of the cells and 50 % parasitized, a rule on labels only. This yields 5 550 hold-out cells from 40 patients (28 infected; 50.7 % parasitized). The remaining 160 patients are split into ten folds of 15–17 patients by `StratifiedGroupKFold`. The split is committed to the repository and every training run verifies its fingerprint; A1 and A2 use the identical split, and A3 reuses A0's split exactly, so both pairs compare the same cells fold by fold.

YUV normalisation (A2, A3). Following [4], each image is converted to YUV, the luma channel is histogram-equalised, and the result is converted back to RGB, at the image's native size and before the unchanged resize and augmentation. Chroma is untouched and the black background stays black. We omit the 120×120 resize and the unspecified edge kernel of [4]. Fig. 1 shows the effect: the narrow brightness range of

each cell is stretched over the full range, which also amplifies texture.

C. Statistical analysis

All decisions below were committed to the repository before the A1–A3 models were trained. Patients are the unit of independence, so every confidence interval comes from a cluster bootstrap that resamples whole patients (10 000 resamples). The primary metric is the Matthews correlation coefficient (MCC), which is insensitive to the class mix that varies between patient folds; accuracy, recall (sensitivity for parasitized cells), specificity, balanced accuracy, F_2 and ROC-AUC are reported as well. The primary protocol evaluates clean (un-augmented) images; the paper's augmented protocol is reported alongside. Paired contrasts (A2 vs A1, A3 vs A0) use the Nadeau–Bengio corrected resampled t-test on the ten per-fold differences [14]—a plain paired t-test is too optimistic because the folds' training sets overlap by about 89 %—and a paired patient bootstrap of the ensemble's hold-out difference. A1 and A0 have different hold-out sets, so their ensembles are compared with independent patient bootstraps, and their folds with a Welch-type t-test in which each arm's variance carries the same Nadeau–Bengio correction. (Our plan used a plain Welch test here; it treated each arm's folds as independent and overstated the evidence, so we corrected it and report both.) p-values are Holm-corrected over the three contrasts [15]. The decision rules restate the hypotheses as registered in our Phase 2 plan: H1 is supported if A1's ensemble MCC is lower than A0's with a 95 % interval excluding zero; H2 is rejected if A2's mean fold-MCC gain over A1 is smaller than A1's fold-to-fold standard deviation; the control shows "little change" if $|\mathrm{A3}-\mathrm{A0}|$ is below A0's fold standard deviation.

Exploratory analyses. After the results, a dated addendum fixed four further analyses before any of them was run:

- *Equivalence.* Two one-sided tests [16] check equivalence within ± 1 pp of MCC for the YUV contrasts, and we report each contrast's minimum detectable effect at 80 % power.
- *Leakage on identical cells.* A0's and A1's ensembles are scored on the 1 095 A0 hold-out cells that belong to A1's 40 held-out patients. The two cross-validations' out-of-fold predictions are compared on the 17 591 cells both contain. Each A0 model had seen other cells of the patient; no A1 model had.
- *Patient-level diagnosis.* A rule calls a patient positive if at least τ cells reach $P(\text{parasitized}) \geq t$. It is chosen on the 160 cross-validation patients (highest Youden J with at least 95 % specificity) and applied once to the 40 held-out patients.
- *Error audit.* We test whether errors cluster by patient and flag candidate label errors by confident learning [17].

V. Results

A. Reproduction

Table II compares our ten-fold reproduction with the outputs printed by the authors' code. The fold models match the paper almost exactly (97.55 vs. 97.56 % mean accuracy on their cross-validation test parts), and every headline metric is within 1 pp. The ensemble is 0.67 pp short of the paper's 98.29 %: our ensemble gains 0.25 pp over its average member where the paper's gained 0.60 pp. The gap is about three standard errors of a 5 512-cell test set; the paper's unrecoverable split and the newer TensorFlow are the remaining differences. Scoring clean instead of augmented images changes our numbers by at most 0.23 pp, so the paper's augmented evaluation does not explain its results. Reading the authors' code also revealed that the paper's text swaps the ensemble's precision and recall (the true recall is 97.74 %, not 98.82 %), and that its Table 6 "accuracy" is $(\text{precision} + \text{recall})/2$ of each report.

TABLE II
Reproduction of [8] on the 5,512-cell hold-out set (%). "Paper protocol" scores augmented images, as the authors' code did; "clean" scores the original images.

Metric	Paper	Ours (paper)	Ours (clean)	Δ
Mean fold accuracy	97.56	97.55	97.78	−0.01
Ensemble accuracy	98.29	97.62	97.64	−0.67
Ensemble recall	97.74	97.07	97.18	−0.67
Ensemble precision	98.82	98.11	98.04	−0.71
Ensemble specificity	98.84	98.17	98.09	−0.67
Ensemble F1	98.28	97.59	97.61	−0.69
Ensemble ROC-AUC	99.76	99.73	99.68	−0.03

B. H1: patient-level split

Table III lists all four experiments and Table IV the three contrasts. On the 40 held-out patients, the A1 ensemble reaches **98.45 %** accuracy (95 % CI 97.94–98.91) and an MCC of 96.90, no lower than A0's 97.64 % on its image-split hold-out set (MCC difference $+1.62$ pp, 95 % CI $+0.01$ to $+3.12$, Holm $p = 0.15$). By the pre-registered rule, *H1 is not supported*: the paper's accuracy largely survives unseen patients.

Cross-validation tells a more careful story (Fig. 4). Its ten test parts together cover 160 patients, a broader test than one draw of 40. There the patient split lowers the mean fold MCC from 95.57 to 93.92, a point estimate of -1.65 pp (95 % CI -3.28 to -0.02, Holm $p = 0.14$). It lowers the mean fold accuracy from 97.78 to 97.24 % and widens the fold-to-fold spread 2.5-fold (MCC SD 0.60 to 1.51), as H1 predicted. The uncorrected Welch test of our plan gave $p = 0.0075$ (Holm 0.023). The same A1 models score 98.22 % on the 40 held-out patients but 97.24 % in cross-validation: this particular hold-out draw is about 1 pp easier than an average group of unseen patients. Validation on unseen patients also stops improving earlier: the best epoch falls at a median of 17 instead of 28.5.

Leakage on identical cells. Both comparisons above score different cells, and the unpaired hold-out comparison could detect only about 2.2 pp of MCC with 80 % power. Scoring the same cells removes both problems (Fig. 3, Table VI).

- On the 17 591 cross-validation cells both arms share, A1's out-of-fold models score 0.85 pp lower in MCC than A0's (95 % CI 0.05–2.02; accuracy -0.43 pp).

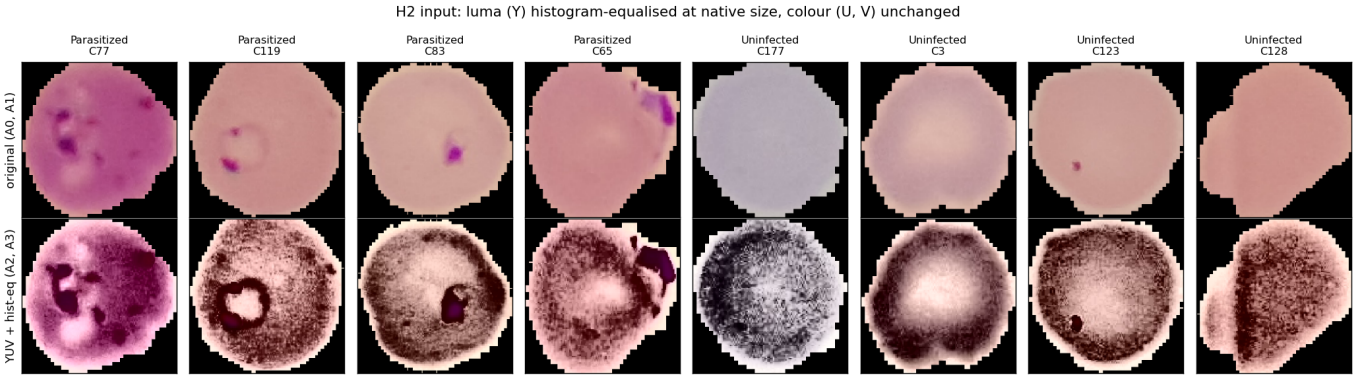
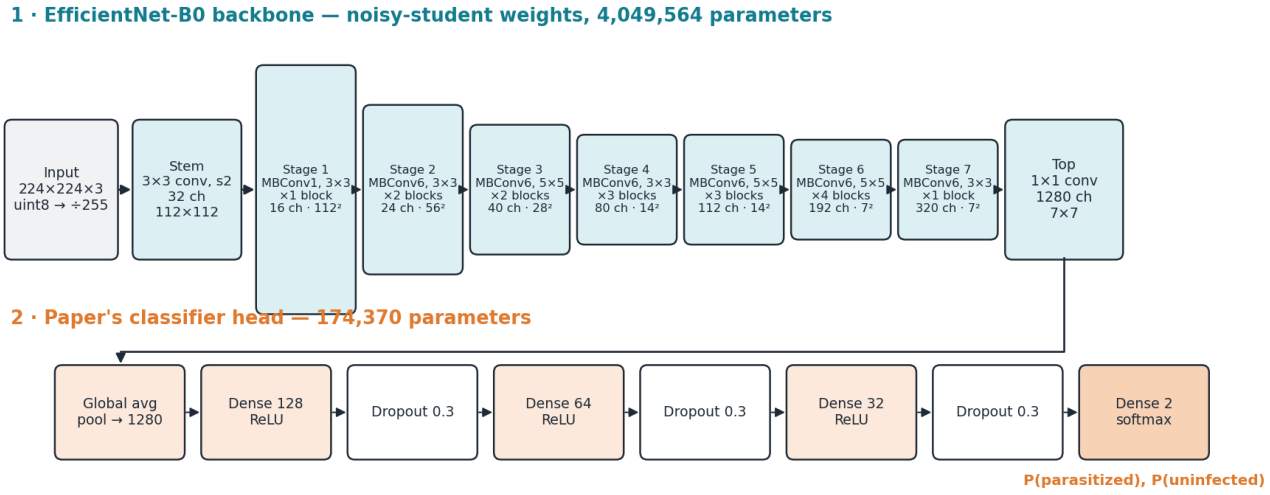


Fig. 1. Input to H2. Top: original cells from eight different patients. Bottom: after YUV conversion and luma histogram equalisation.



Resolution shrinks 224 → 112 → 56 → 28 → 14 → 7 while channels grow 3 → 32 → … → 320 → 1280: the network trades 'where' for 'what' (box height ∝ resolution).
16 MBConv blocks in 7 stages · 81 convolution layers (65 normal + 16 depthwise) · 49 batch-norm layers · 230 Keras layers in the backbone · 4 dense layers in the head.

Total 4,223,934 parameters = 4,181,918 trainable + 42,016 non-trainable (batch-norm running statistics) — exactly the paper's Table 2.

Fig. 2. The reproduced model: noisy-student EfficientNet-B0 (stem, 16 MBConv blocks in seven stages, 1×1 top convolution) and the paper's dense head. 4,223,934 parameters, exactly the paper's Table 2.

- On the 1 095 A0 hold-out cells of A1's 40 held-out patients, the two ensembles have identical confusion matrices (difference 0.00 pp, 95 % CI -0.90 to $+0.98$), and 19 of each ensemble's 22 errors fall on the same cells.

Patient leakage is therefore real on this dataset, but small. A0's models also trained on more patients, so these differences, if anything, overstate it.

C. H2: YUV stain normalisation

On the same unseen patients, adding YUV normalisation (A2 vs. A1) changes the mean fold MCC by -0.60 pp, which is smaller than A1's fold-to-fold standard deviation of 1.51 pp. The pre-registered rule therefore rejects H2, and the corrected resampled t-test agrees ($p = 0.13$). On the hold-out set the ensemble MCC changes by -0.26 pp (95 % CI -0.85 to $+0.23$); recall is unchanged ($+0.18$ pp) and specificity is slightly lower (-0.44 pp, not significant after correction) (Fig. 5). The control A3 vs. A0 shows no detectable effect when patients are shared (ensemble MCC $+0.00$ pp, 95 % CI -0.52 to $+0.53$). The analysis under the paper's augmented protocol gives the same three verdicts.

Equivalence tests bound the effect (Table VI).

- On the hold-out set, YUV changes MCC by -0.73 to $+0.16$ pp (90 % CI). That lies inside a ± 1 pp margin (TOST $p = 0.008$) and below the 0.77 pp this paired design can detect.
- At the fold level the interval (-1.27 to $+0.06$) is too wide to establish equivalence.
- The control is equivalent within ± 1 pp at both levels.

TABLE III

ABLATION RESULTS, CLEAN EVALUATION (%). FOLD COLUMNS: MEAN ± SD OVER THE TEN CV TEST PARTS. ENSEMBLE COLUMNS: THE TEN-MODEL ENSEMBLE ON EACH HOLD-OUT SET, WITH 95 % PATIENT-BOOTSTRAP INTERVALS. MCC ×100.

ID	Split	Prep.	Hold-out cells / patients	Fold MCC	Fold accuracy	Ens. accuracy	Ens. MCC	Ens. recall
A0	image	none	5,512 / 200	95.57 ± 0.60	97.78 ± 0.30	97.64 [97.02, 98.17]	95.29 [94.03, 96.32]	97.18 [96.03, 98.06]
A1	patient	none	5,550 / 40	93.92 ± 1.51	97.24 ± 0.66	98.45 [97.94, 98.91]	96.90 [95.70, 97.79]	98.05 [96.79, 98.85]
A2	patient	YUV	5,550 / 40	93.32 ± 1.66	96.96 ± 0.75	98.32 [97.71, 98.86]	96.65 [95.23, 97.68]	98.22 [97.02, 99.02]
A3	image	YUV	5,512 / 200	95.20 ± 0.37	97.60 ± 0.19	97.64 [97.02, 98.17]	95.29 [94.04, 96.32]	97.14 [95.94, 98.05]

TABLE IV

CONTRASTS (NEW − REFERENCE, PERCENTAGE POINTS). FOLD: MEAN DIFFERENCE OVER THE TEN FOLDS WITH 95 % CI (NADEAU–BENGIO CORRECTED t; FOR THE UNPAIRED H1 CONTRAST, A WELCH-TYPE t WITH THE CORRECTION APPLIED TO EACH ARM). ENSEMBLE: HOLD-OUT DIFFERENCE WITH 95 % PATIENT-BOOTSTRAP CI. p-VALUES HOLM-CORRECTED OVER THE THREE CONTRASTS.

Contrast	Metric	Fold Δ [95 % CI]	Fold p_{Holm}	Ensemble Δ [95 % CI]	Ens. p_{Holm}
H1: A1 − A0 (unpaired)	MCC	−1.65 [−3.28, −0.02]	0.142	+1.62 [+0.01, +3.12]	0.146
	Accuracy	−0.54 [−1.27, +0.18]	0.364	+0.81 [+0.06, +1.57]	0.102
	Recall	−1.27 [−2.71, +0.18]	0.240	+0.87 [−0.67, +2.25]	0.748
	Specificity	−0.26 [−1.10, +0.57]	0.687	+0.77 [+0.00, +1.53]	0.118
	ROC-AUC	−0.23 [−0.54, +0.08]	0.180	+0.12 [−0.10, +0.32]	0.593
H2: A2 − A1 (paired)	MCC	−0.60 [−1.43, +0.22]	0.246	−0.26 [−0.85, +0.23]	0.637
	Accuracy	−0.28 [−0.67, +0.11]	0.364	−0.13 [−0.41, +0.11]	0.756
	Recall	−0.20 [−0.61, +0.21]	0.609	+0.18 [−0.11, +0.53]	0.748
	Specificity	−0.33 [−0.91, +0.25]	0.687	−0.44 [−0.92, −0.04]	0.118
	ROC-AUC	−0.18 [−0.39, +0.02]	0.180	−0.04 [−0.09, +0.02]	0.593
Control: A3 − A0 (paired)	MCC	−0.37 [−0.87, +0.12]	0.246	+0.00 [−0.52, +0.53]	1.000
	Accuracy	−0.19 [−0.43, +0.06]	0.364	+0.00 [−0.26, +0.27]	1.000
	Recall	−0.10 [−0.43, +0.23]	0.609	−0.04 [−0.43, +0.33]	0.935
	Specificity	−0.27 [−0.85, +0.30]	0.687	+0.04 [−0.36, +0.43]	0.917
	ROC-AUC	−0.16 [−0.33, +0.01]	0.180	−0.02 [−0.11, +0.05]	0.663

TABLE V

RECALL-FIRST THRESHOLD CHOSEN ON THE POOLED OUT-OF-FOLD CV PREDICTIONS (MAX F_2) AND APPLIED ONCE TO THE HOLD-OUT SET; EXPECTED CALIBRATION ERROR (ECE) OF THE ENSEMBLE. CLEAN EVALUATION.

ID	Threshold	Recall at 0.5 → at threshold	Specificity	ECE
A0	0.192	97.18 → 98.72	95.22	0.053
A1	0.189	98.05 → 99.22	93.93	0.065
A2	0.143	98.22 → 99.22	91.74	0.060
A3	0.159	97.14 → 98.75	94.14	0.052

TABLE VI

EXPLORATORY ANALYSES FIXED IN A DATED ADDENDUM BEFORE THEY WERE RUN (MCC DIFFERENCES IN PP; CLEAN EVALUATION).

Analysis	Estimate	Interval
Leakage on identical cells, A1 − A0 (95 % CI)		
ensembles: 1,095 cells, 40 pts	+0.00	[−0.90, +0.98]
out-of-fold: 17,591 cells, 160 pts	−0.85	[−2.02, −0.05]
YUV effect within ±1 pp? (90 % CI)		
H2 A2 − A1, ensemble	−0.26	[−0.73, +0.16] yes
H2 A2 − A1, folds	−0.60	[−1.27, +0.06] no
control A3 − A0, ensemble	+0.00	[−0.44, +0.44] yes
control A3 − A0, folds	−0.37	[−0.77, +0.03] yes
Patient level, 40 held-out patients (exact 95 % CI)		
sensitivity	26/28	92.9 % [76.5, 99.1]
specificity	12/12	100.0 % [73.5, 100.0]

D. Screening-oriented analysis

For screening, a missed infected cell is costlier than a false alarm. Choosing the decision threshold on the pooled out-of-fold cross-validation predictions (maximum F_2) and applying it once to the hold-out set raises recall to 98.7–99.2 % at 91.7–95.2 % specificity (Table V). The ensembles are slightly under-confident (ECE 0.05–0.07), as the paper's label smoothing predicts.

E. Patient-level diagnosis (exploratory)

The rule chosen on the 160 cross-validation patients' out-of-fold predictions calls a patient positive if *at least 3 cells have* $P(\textit{parasitized}) \geq 0.80$. On those patients its sensitivity is 96.7 % and its specificity 100 %; selecting the rule in a nested leave-one-fold-out gives 96.7 % and 97.4 %. Applied once to the 40 held-out patients with the A1 ensemble, it identifies 26 of 28 infected patients (92.9 %, exact 95 % CI 76.5–99.1) and all 12 healthy ones (100 %, 73.5–100).

The two missed patients, C53 and C73, are the only held-out patients with just three parasitized crops. When each infected patient keeps only one or two parasitized crops, sensitivity falls to 3.6 and 12.3 %. NIH kept about 68 uninfected crops per patient but curated the parasitized ones, so these numbers describe aggregation on curated crops, not diagnostic accuracy on a smear.

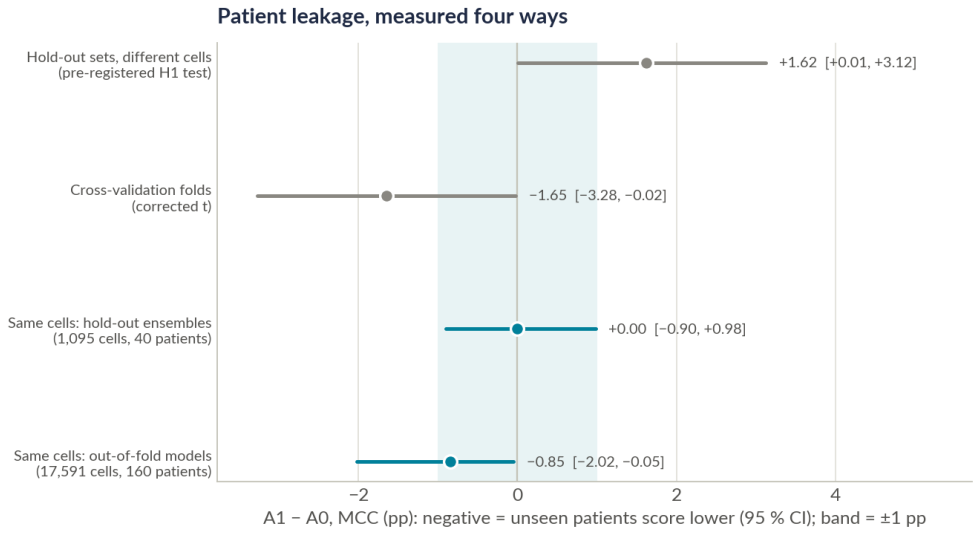


Fig. 3. A1 − A0 in MCC, measured four ways (95 % CIs; shaded: ±1 pp). The pre-registered hold-out comparison scores different cells; the cross-validation folds use the corrected t-test; the two teal rows score identical cells.

F. Error audit (exploratory)

- *Errors cluster by patient* (permutation test $p < 0.001$): half of A1's 86 hold-out errors come from 7 of the 40 patients, and 14 patients have none.
- *False alarms concentrate in infected patients' smears.* The model flags 1.57 % of their uninfected-labelled cells, against 0.12 % in healthy patients (difference 1.45 pp, 95 % CI 0.82–2.14); out of fold the rates are 2.35 and 0.85 %. This fits parasites missed in the labels, or genuinely ambiguous cells next to parasites; without an expert we cannot tell which.
- *Candidate label errors.* Confident learning flags 8 hold-out cells. Removing them leaves both YUV contrasts unchanged.

G. Compute

Each of the 40 models trains for 33 epochs in about 11 minutes on an RTX 4090 (10.2–12.8 minutes for A1–A3) and in 11–18 minutes on an L40S, against 4 h 46 min per fold reported for a GTX 1050. A1–A3 ran as one cloud pod per fold that trains the three experiments in turn: 30 models in about one hour of wall-clock time for US$6.95. At batch size 16 a training step takes about 16 ms and is dominated by per-step overhead rather than arithmetic. The data-centre L40S was slower than the consumer RTX 4090 even with twice the CPU cores, so larger GPUs are unlikely to shorten training much without changing the recipe.

VI. Discussion

Why is the leakage effect small here? Rajaraman *et al.* [2] reported a drop from 98.6 to 95.9 % when moving from cell-level to patient-level evaluation with features from pre-trained networks. Our end-to-end fine-tuned ensemble loses about half a point of accuracy: 0.43 pp on identical cells (0.85 pp of MCC) and 0.54 pp in cross-validation. Two plausible reasons, which our experiments do not separate, are that the decisive cue (stained parasite chromatin) looks alike across patients, and that the paper's heavy colour, contrast and blur augmentation already prevents the network from relying on a patient's staining.

One hold-out set is not enough. The 40-patient hold-out set is about 1 pp easier than an average fold of unseen patients. That is larger than the leakage effect and larger than the paper's own margins over competing methods. Patient clustering also inflates the variance of hold-out accuracy 2.3-fold (MCC 2.6-fold) compared with treating cells as independent. Sub-1-pp differences between published methods therefore cannot be resolved on a single random hold-out set. The minimum needed is cross-validation over patients with patient-level confidence intervals, and paired comparisons on identical cells where possible, as we do here.

Why does YUV normalisation not help? Umer *et al.* [4] reported 99.96 % with their full preprocessing against 49.61 % without it. That comparison is against a model that did not train, not against a strong baseline. With a pre-trained network and an augmentation pipeline that already randomises brightness and contrast (including CLAHE), explicit equalisation adds nothing. The step also changes luma only: it normalises brightness, not stain colour. In histopathology, colour augmentation rather than normalisation gave the most robust models across laboratories [18], and the recipe here has none, so a colour-augmentation arm tested on another laboratory's smears is the natural next experiment. By stretching each cell's narrow luma range to the full scale (Fig. 1), the step also amplifies texture and noise, consistent with the slightly lower specificity we observe.

VII. Limitations

The patient-level analysis is exploratory. NIH kept about 68 uninfected crops per patient but curated the parasitized ones,

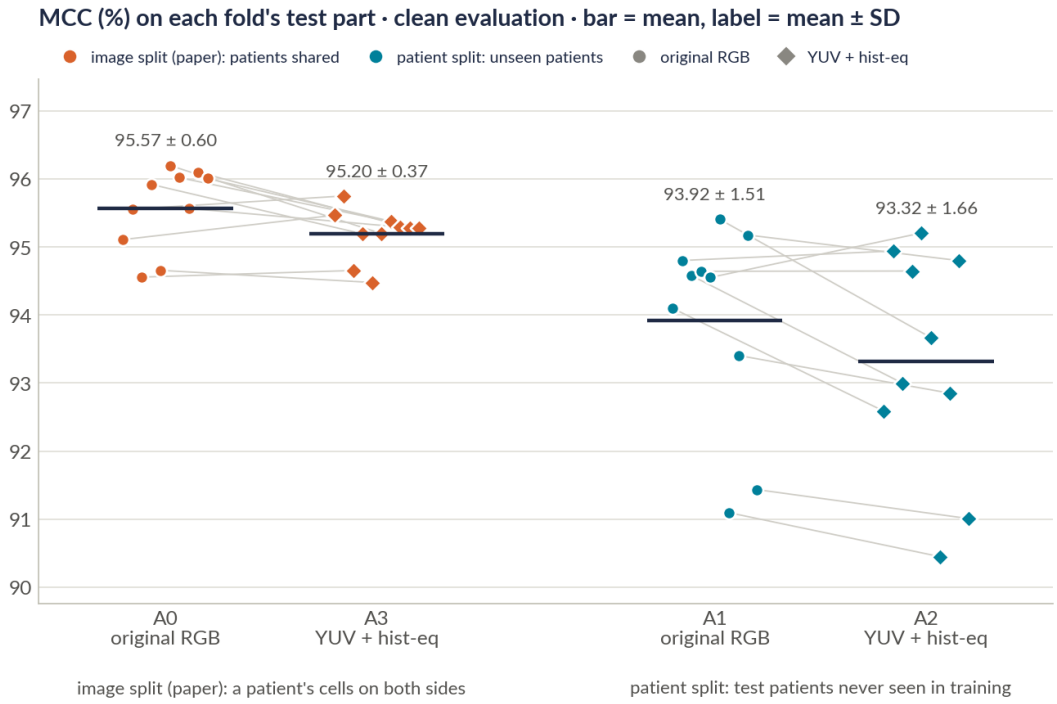


Fig. 4. MCC of every fold model on its cross-validation test part (clean evaluation). Orange: the paper's image split; teal: the patient split. Grey lines join the same fold across the paired YUV change. Patient-split folds score lower and vary more; YUV changes little.

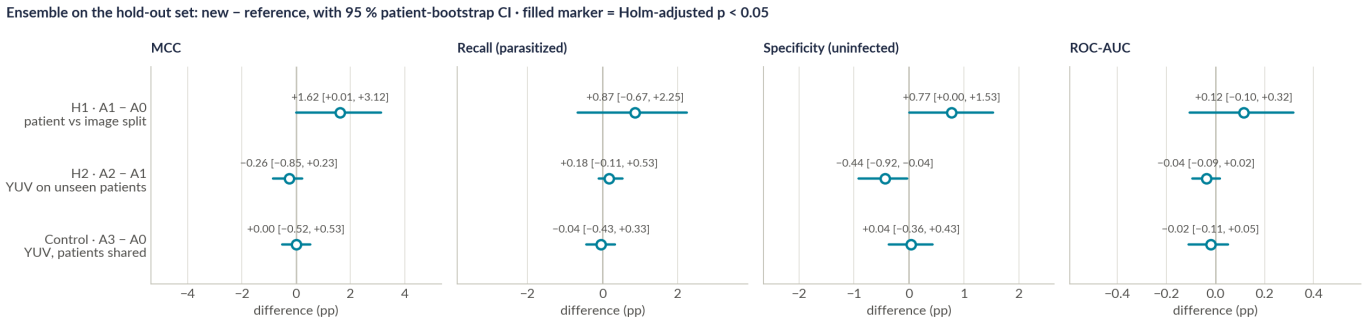


Fig. 5. The three contrasts on the ensembles' hold-out sets, with 95 % patient-bootstrap intervals. None survives the Holm correction.

so it cannot show diagnostic accuracy on a real smear, and only 12 held-out patients are healthy. All data come from one hospital, so a patient split removes patient leakage but not site or staining-protocol effects. Without an external test set, H2 is tested on unseen patients of the same laboratory, not on another laboratory's stain. Labels come from a single expert. Fuhad *et al.* [5] set aside 1,397 cells (5.1 %) as mislabelled or suspicious, but their list is no longer available, so we keep the original labels, as the paper did. Each comparison rests on one hold-out draw and one training run per fold and arm. The out-of-fold same-cell comparison uses single models, so its interval omits run-to-run noise. A1–A3 ran on RTX 4090 and L40S GPUs (A0 on RTX 4090 only); within a fold, A1, A2 and A3 shared one GPU, so the paired comparisons use identical hardware.

VIII. Conclusion

Rebuilt from its authors' code, the EfficientNet-B0 ensemble of Marques *et al.* reproduces within 1 pp. Its accuracy largely survives patients it has never seen (98.45 % on 40 held-out patients). Measured on identical cells, patient leakage is real but small, under about 2 pp of MCC. YUV histogram-equalisation stain normalisation, reported as highly effective in the literature, does not improve this strong augmented baseline: its effect on unseen patients lies within ±1 pp of MCC. Combining cell calls per patient, a rule chosen without the test patients identifies 26 of 28 infected and all 12 healthy held-out patients, though only on curated crops. Future work should:

- test on an external laboratory's smears;
- re-read the remaining errors against an independent relabelling of the dataset;

- evaluate patient-level rules on whole smears;
- distil the ensemble into a single model light enough for point-of-care use.

Code, results, the analysis plan and all trained models are public in our repository.

REFERENCES

[1] Z. Liang *et al.*, "CNN-based image analysis for malaria diagnosis," in *Proc. IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, 2016.

[2] S. Rajaraman *et al.*, "Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images," *PeerJ*, vol. 6, p. e4568, 2018.

[3] S. Rajaraman, S. Jaeger, and S. K. Antani, "Performance evaluation of deep neural ensembles toward malaria parasite detection in thin-blood smear images," *PeerJ*, vol. 7, p. e6977, 2019.

[4] M. Umer, S. Sadiq, M. Ahmad, S. Ullah, G. S. Choi, and A. Mehmood, "A novel stacked CNN for malarial parasite detection in thin blood smear images," *IEEE Access*, vol. 8, 2020, doi: 10.1109/ACCESS.2020.2994810.

[5] K. M. F. Fuhad *et al.*, "Deep learning based automatic malaria parasite detection from blood smear and its smartphone based application," *Diagnostics*, vol. 10, no. 5, p. 329, 2020.

[6] O. S. Zhao *et al.*, "Convolutional neural networks to automate the screening of malaria in low-resource countries," *PeerJ*, vol. 8, p. e9674, 2020.

[7] M. Masud *et al.*, "Leveraging deep learning techniques for malaria parasite detection using mobile application," *Wireless Commun. Mobile Comput.*, vol. 2020, p. 8895429, 2020.

[8] G. Marques, A. Ferreras, and I. de la Torre-Díez, "An ensemble-based approach for automated medical diagnosis of malaria using EfficientNet," *Multimedia Tools Appl.*, vol. 81, pp. 28061–28078, 2022, doi: 10.1007/s11042-022-12624-6.

[9] G. Madhu *et al.*, "Intelligent diagnostic model for malaria parasite detection and classification using imperative inception-based capsule neural networks," *Sci. Rep.*, vol. 13, Art. no. 13377, 2023.

[10] K. Hemachandran *et al.*, "Performance analysis of deep learning algorithms in diagnosis of malaria disease," *Diagnostics*, vol. 13, no. 3, p. 534, 2023.

[11] M. Mujahid *et al.*, "Efficient deep learning-based approach for malaria detection using red blood cell smears," *Sci. Rep.*, vol. 14, Art. no. 13249, 2024.

[12] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.

[13] Q. Xie, M.-T. Luong, E. Hovy, and Q. V. Le, "Self-training with noisy student improves ImageNet classification," in *Proc. CVPR*, 2020, pp. 10687–10698.

[14] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Mach. Learn.*, vol. 52, no. 3, pp. 239–281, 2003.

[15] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[16] D. Lakens, "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses," *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017.

[17] C. G. Northcutt, L. Jiang, and I. L. Chuang, "Confident learning: Estimating uncertainty in dataset labels," *J. Artif. Intell. Res.*, vol. 70, pp. 1373–1411, 2021.

[18] D. Tellez *et al.*, "Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology," *Med. Image Anal.*, vol. 58, Art. no. 101544, 2019.